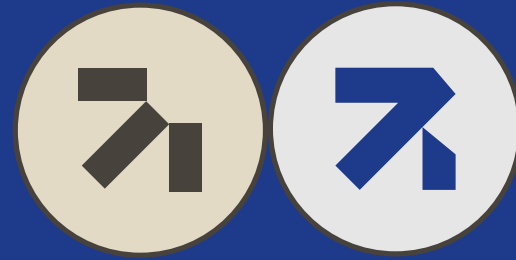
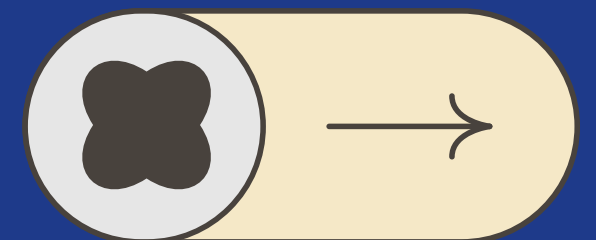


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27/09/2026



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IBM CAPSTONE PROJECT



PROJECT OBJECTIVES

Summary of global developer profiles, current technology stack adoption, and future state of the global developer community from the Stack Overflow Developer Survey

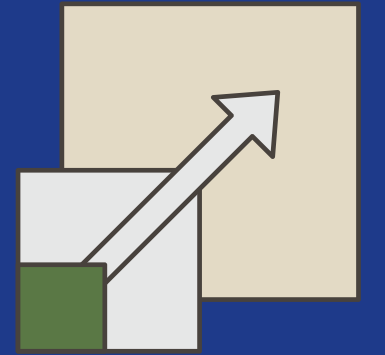
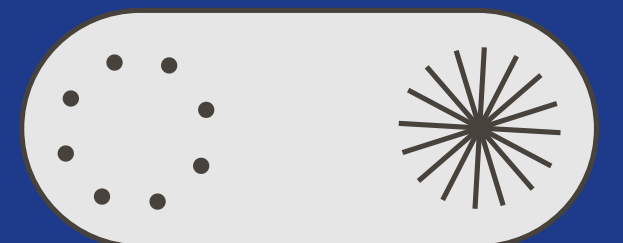
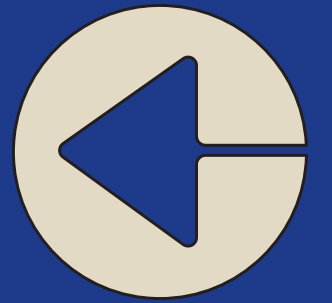


TABLE OF CONTENTS

- Executive Summary
- Introduction
- Methodology
- Results
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- Conclusion

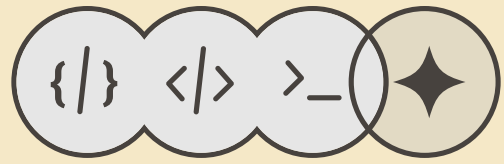


EXECUTIVE SUMMARY

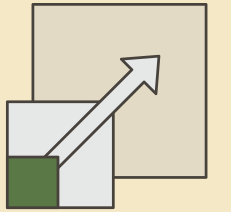


- **Tech Stack Continuity:** High alignment between production tools and desired technologies indicates strong ecosystem stability across:
 - Tools, frameworks, platforms, and databases
- **Shift Toward Systems Efficiency:** Learning trends demonstrate a strategic pivot toward modern systems languages (e.g., Rust, Go)
- **Near-monopoly** of PostgreSQL since it is the dominant database across all business sizes
- **Generational Divide in Web Frameworks:**
 - Senior developers (~12 yrs exp) maintain established enterprise stacks (ASP.NET, jQuery)
 - Junior developers (<8 yrs exp) lean toward modern JS/Python frameworks (Next.js, FastAPI)





INTRODUCTION



My partially “personal” project (bc I follow most of the course guidelines). Anyway, in this project, I aim to apply what I’ve learned to real-world survey data, share my findings, and gain personal experience.

Target audience:

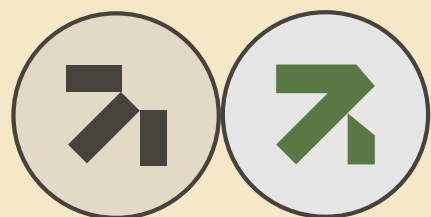
- Peer DA learners
- People with little technology knowledge

Takeaways:

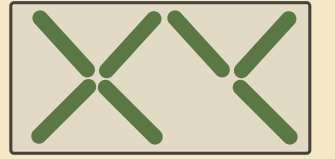
- Industry understanding
- Future trends prediction

Upcoming:

- Methodology
- Analyzing global developer survey data, mapping current technology adoption against future usage desires
- The baseline demographic, age, and educational profiles



METHODOLOGY



1.1 Data storage

- The core dataset (survey_results_updated.csv) was extracted, cleaned, and processed using Python (Pandas) to enable fast, accurate aggregation in Google Looker Studio.

Since there are columns, for example:
LanguageHaveWorkedWith must be `.str.split(";").explode()` to analyze.

Exploding all at once will increase the dataset significantly. Therefore, my solution is that each visualization will have its own table/csv file
→ In total, I've separated the original dataset into 9 smaller csv files

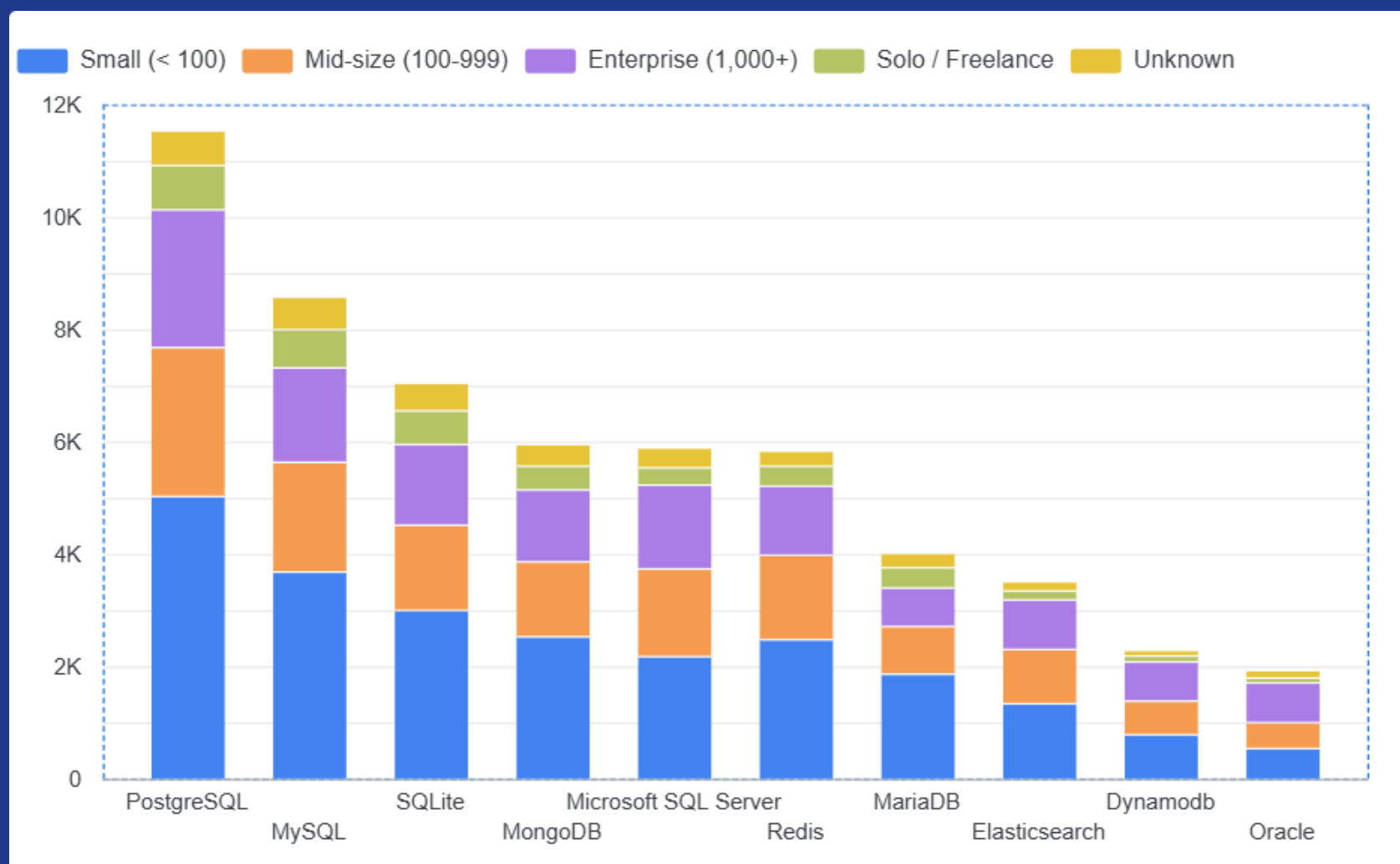




METHODOLOGY

1.2 Classification

- This part is specifically for YearsCodePro, OrgSize columns to **create 2nd dimension** of stacked bar and column charts and better visualization



TOP 10 DATABASES USED
(stacked by OrgSize)

Classifies YearsCodePro into tiers:

- Unknown
- Less than 1 year
- More than 50 years
- < 2 years
- 3-5 years
- 6-10 years
- 11-20 years
- 20+ years

Classifies OrgSize into company-size tiers:

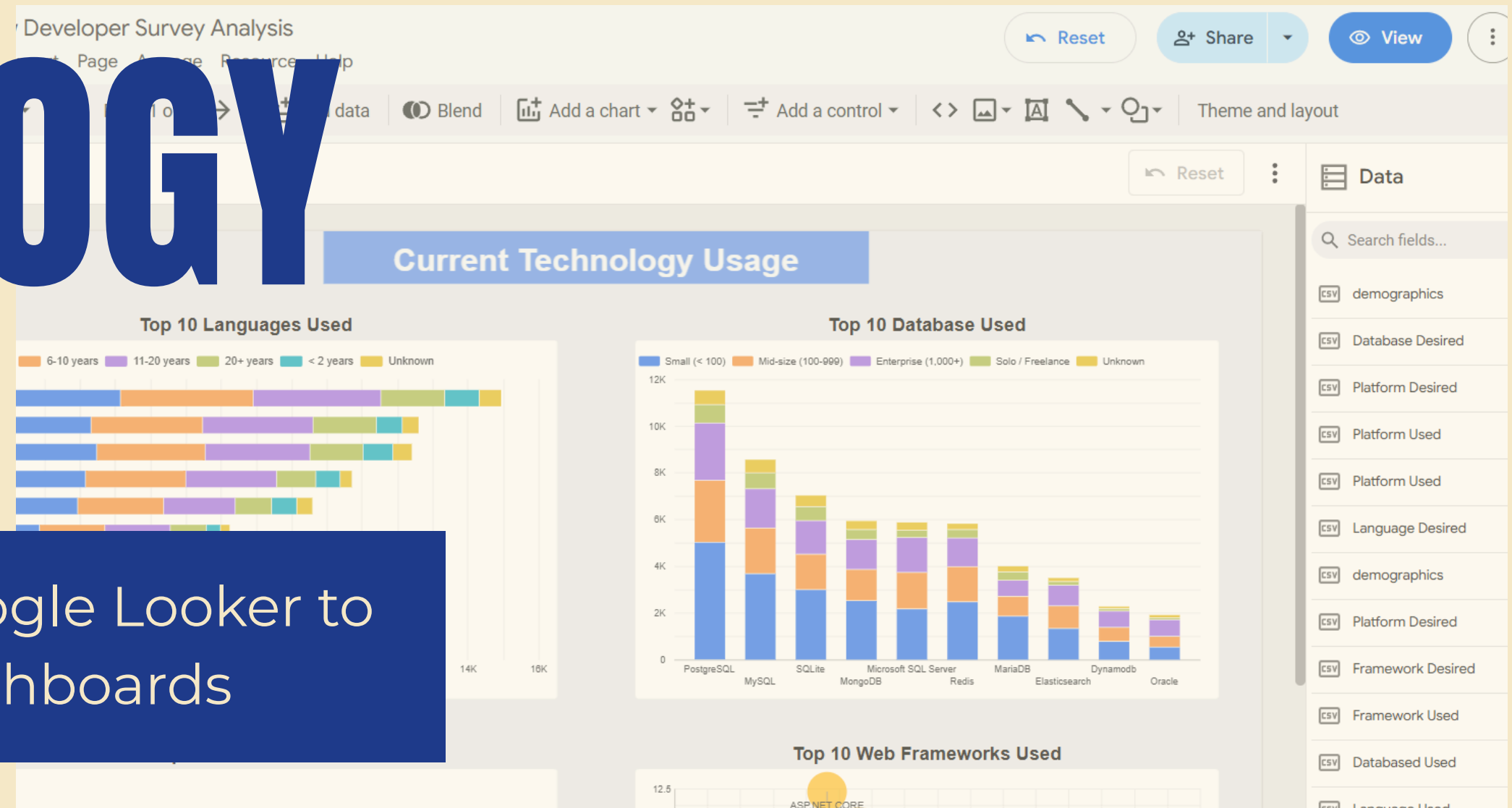
- Unknown
- Solo / Freelance
- Small (< 100)
- Mid-size (100-999)
- Enterprise (1,000+)

Also, “Unknown” class at this stage already handles missing values in the dataset



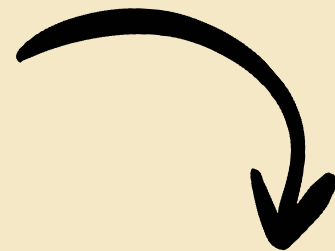
METHODOLOGY

2 Create Dashboard

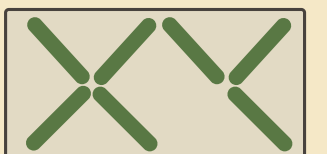


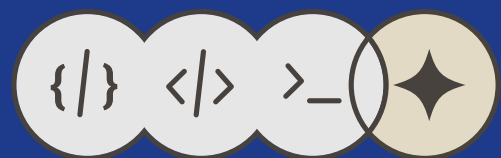
- Upload the extracted csv files to Google Looker to perform aggregation and create dashboards

DASHBOARD LINK

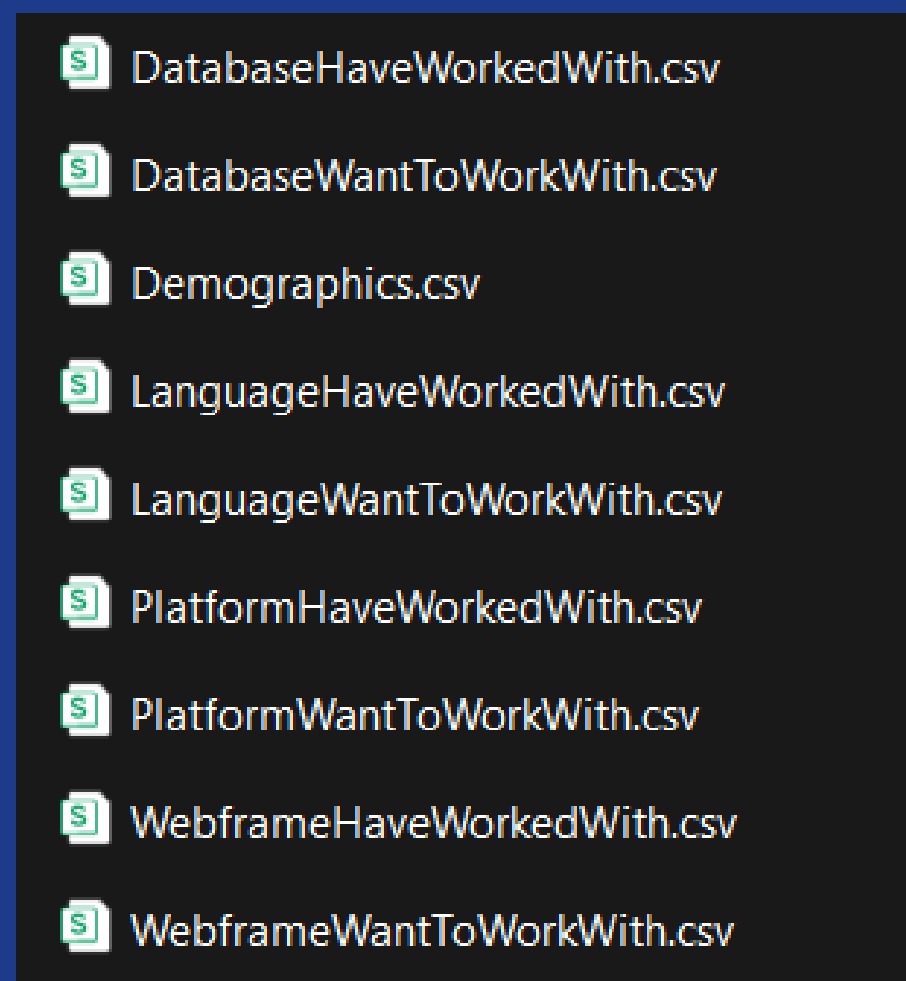
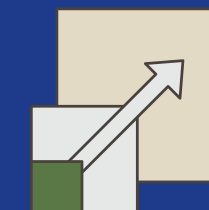


<https://datastudio.google.com/reporting/39b0a36c-e439-4149-9690-8613798ff677>

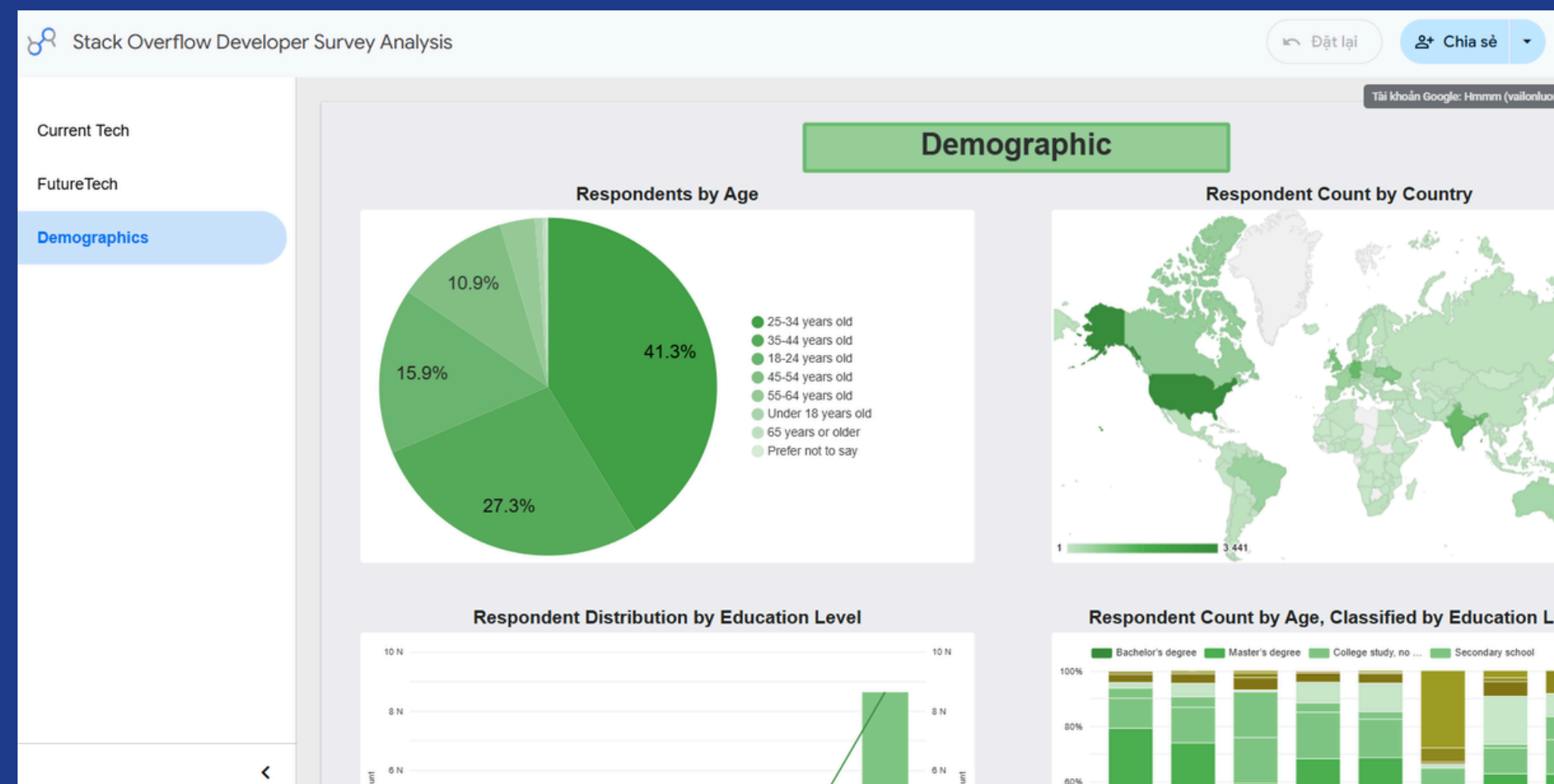




RESULTS



Extracted tables/csv files

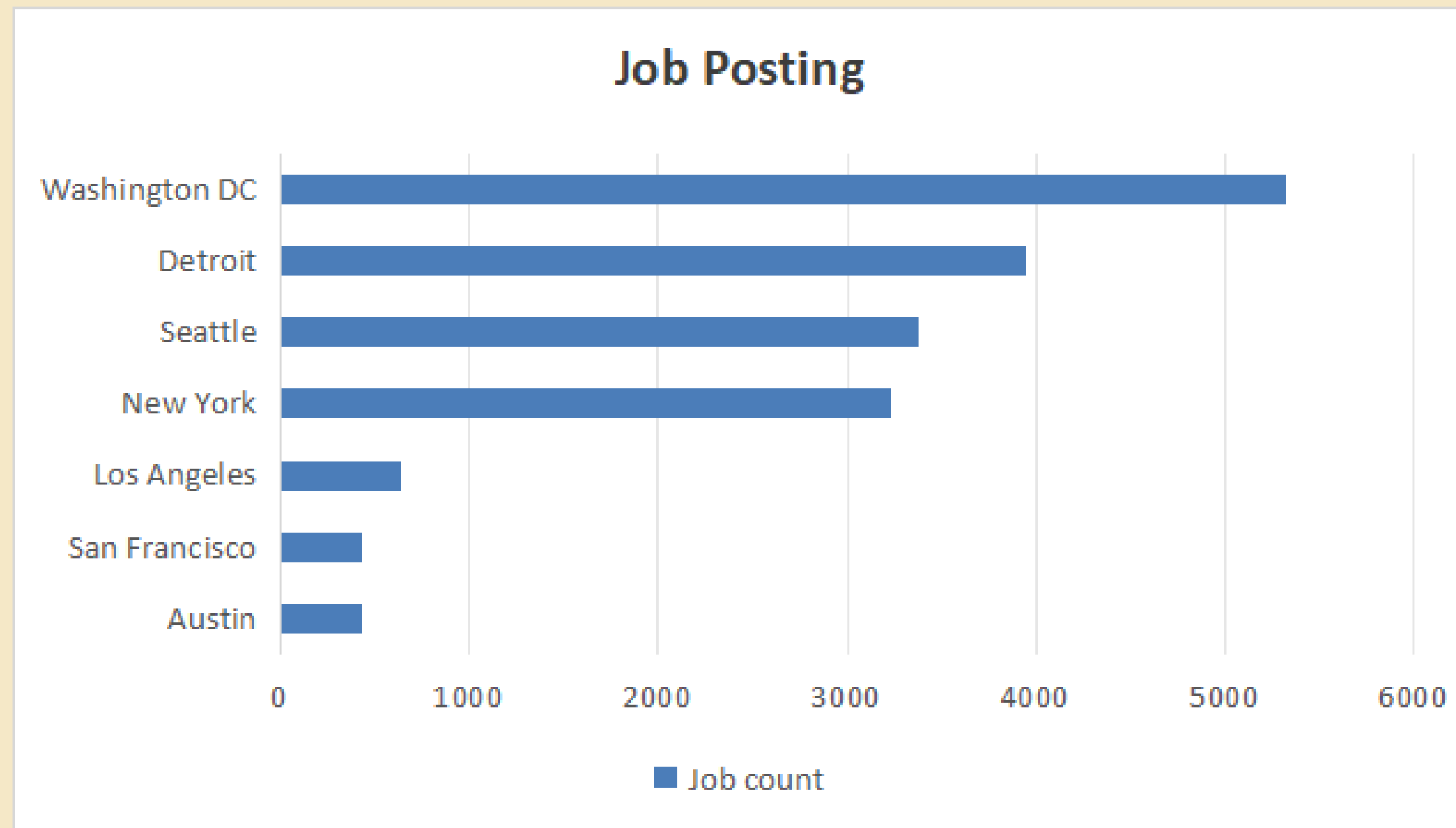


Shareable dashboards via Google Looker Studio

AND PERSONAL EXPERIENCE 

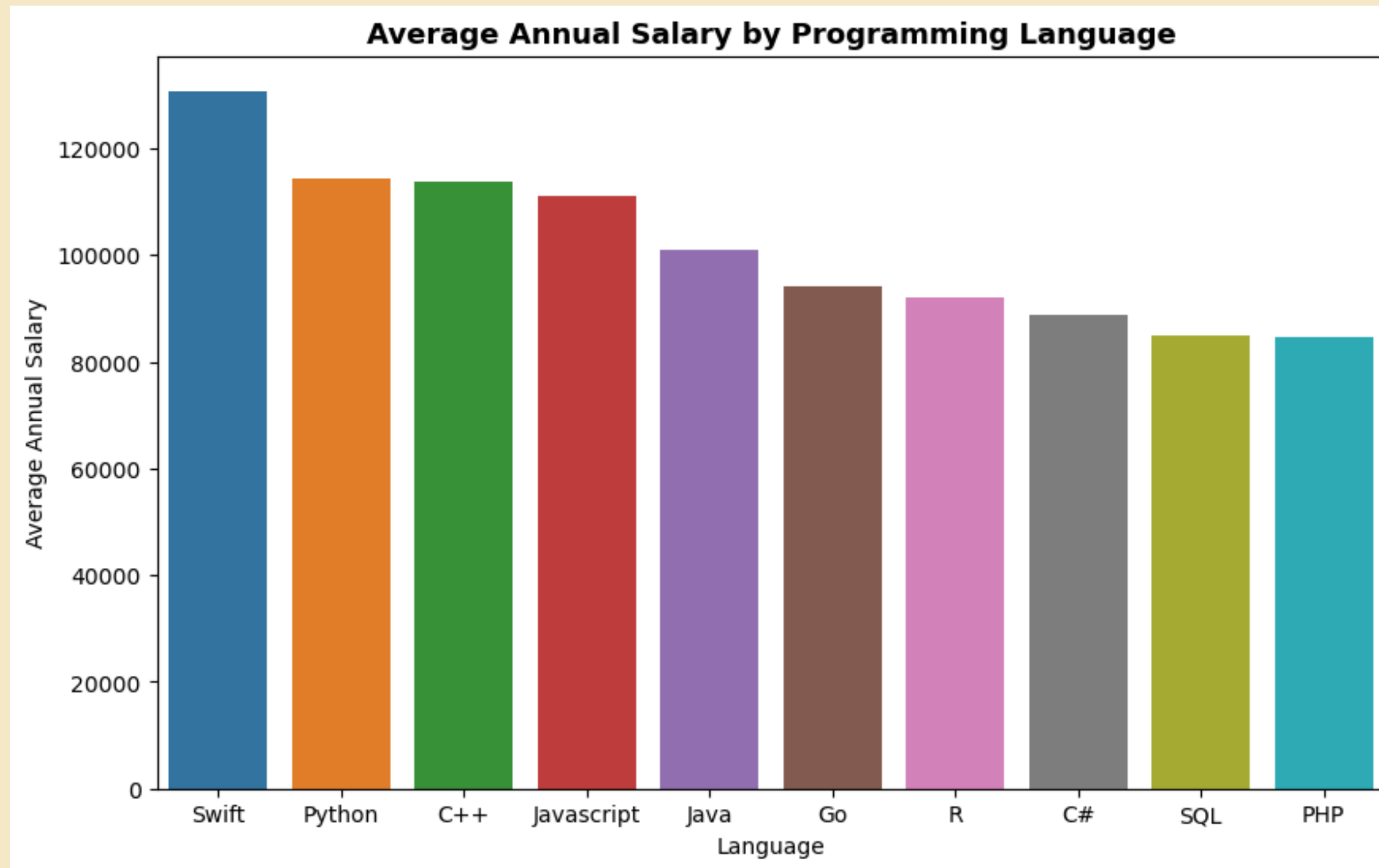
JOB POSTINGS

“job-postings.xlsx” from module 1 visualization



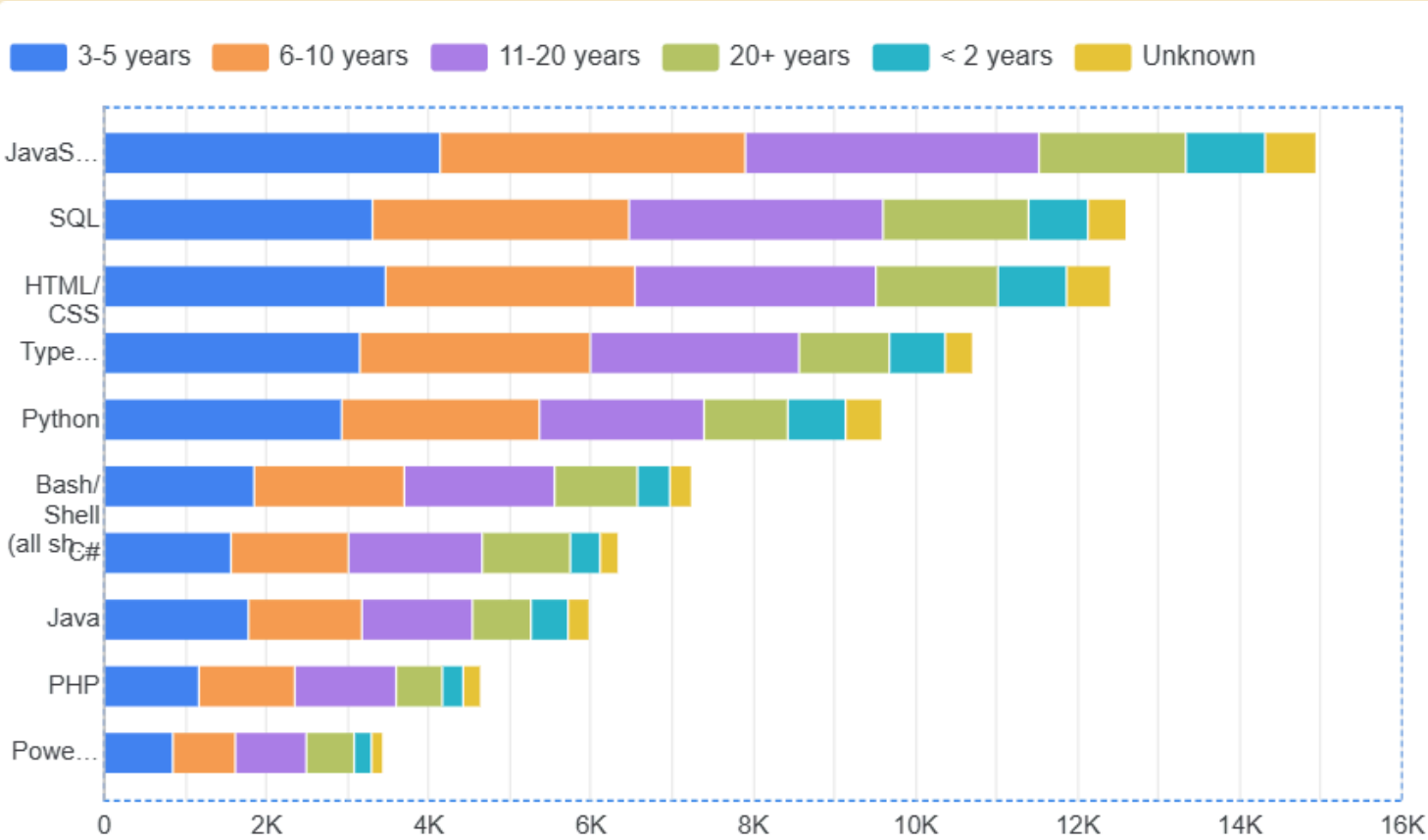
POPULAR LANGUAGES

“popular-languages.csv” from module 1 visualization



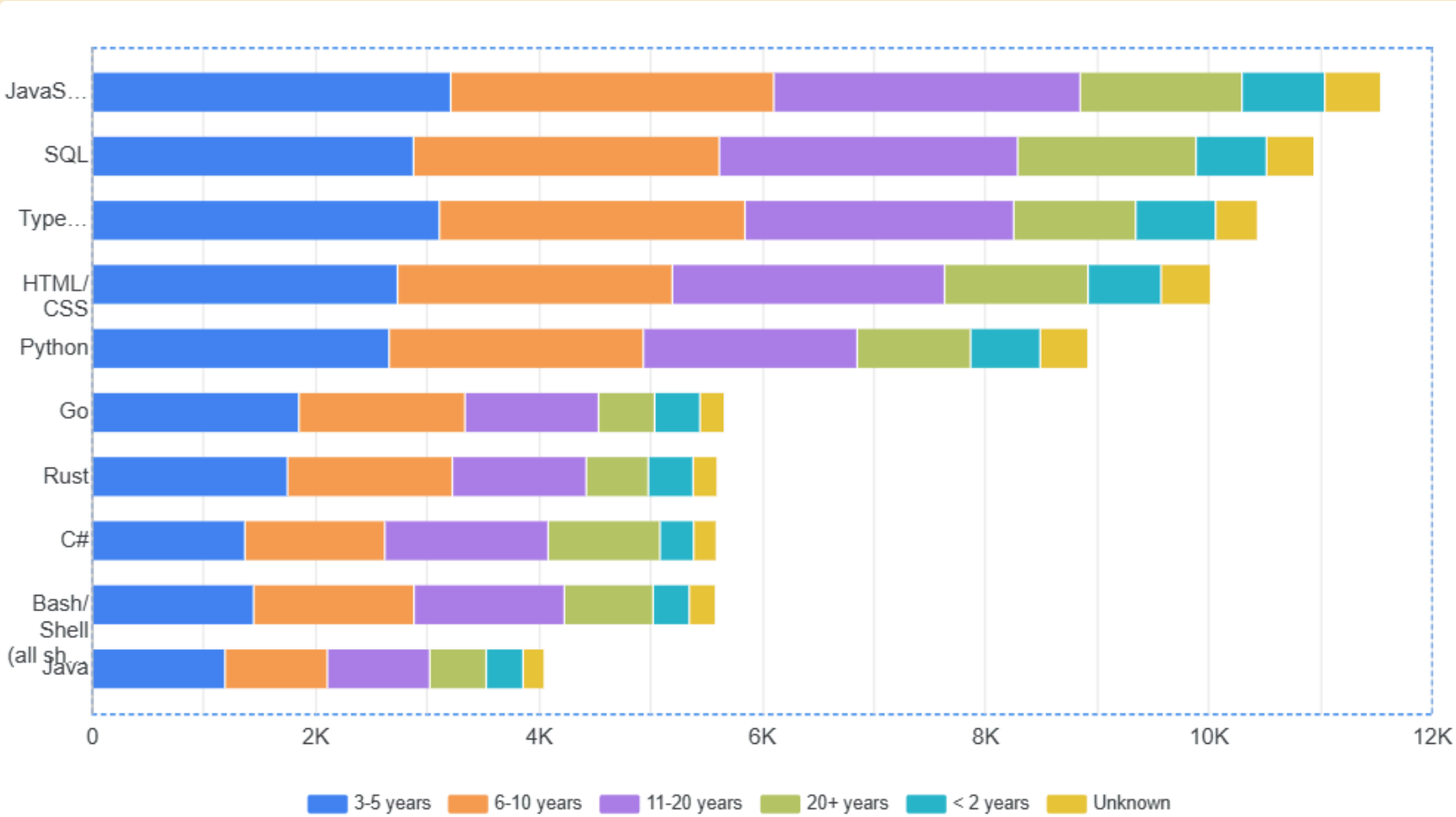
PROGRAMMING LANGUAGE TRENDS

CURRENT YEAR



TOP 10 LANGUAGE USED

NEXT YEAR



TOP 10 LANGUAGE DESIRED

Both have similar in Language choice to learn and YearsCodePro distribution

PROGRAMMING LANGUAGE TRENDS – FINDINGS & IMPLICATIONS

Findings

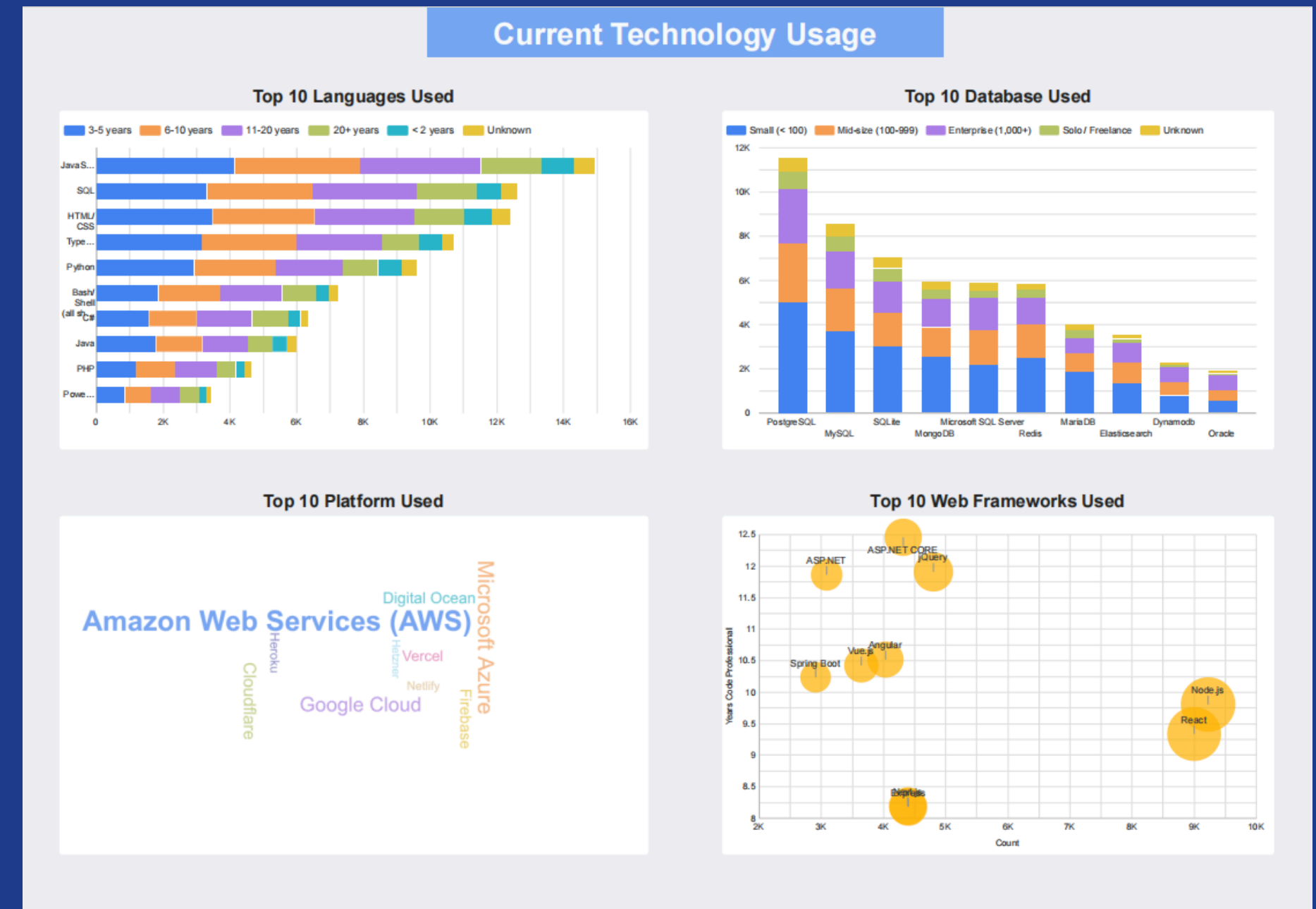
- JavaScript, SQL, TypeScript, HTML/CSS, Python remain the core languages across both
- Rise of systems languages, such as Go, Rust, in the future trend
- Consistent Experience distribution: 3-10 years of experience make up the largest proportion across both

Implications

- Codebase Quality & Efficiency: The growing developer migration toward TypeScript, Go, and Rust reflects an industry-wide shift toward type safety, memory security, and runtime execution efficiency.

DASHBOARD 1: CURRENT TECH USAGE

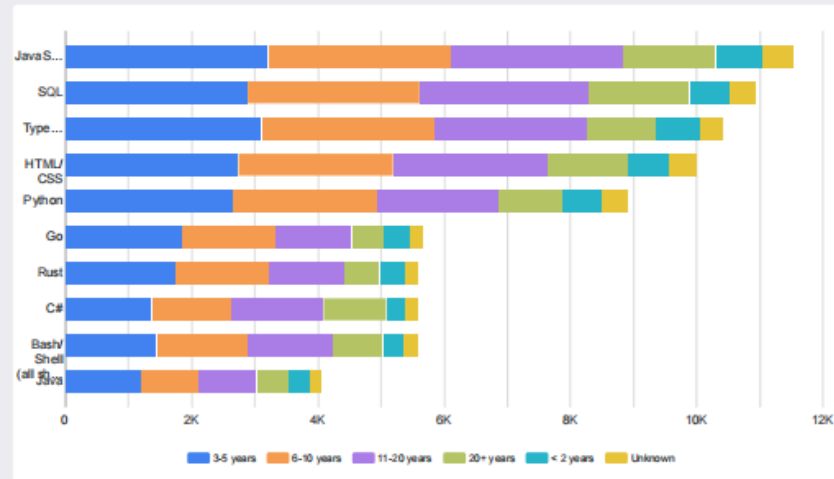
- JavaScript (~15,000), SQL (~12,500), and HTML/CSS (~12,000) form the primary operational language stack for devs with 3-10 years of experience
- Microsoft SQL Server and Oracle skew toward large organizations (100+ employees), whereas the top databases have higher proportional usage in small companies and solo/freelance
- React and Node.js lead overall adoption volume (~9,000–9,500) among developers
- ASP.NET CORE, ASP.NET, and jQuery are maintained by more senior engineers



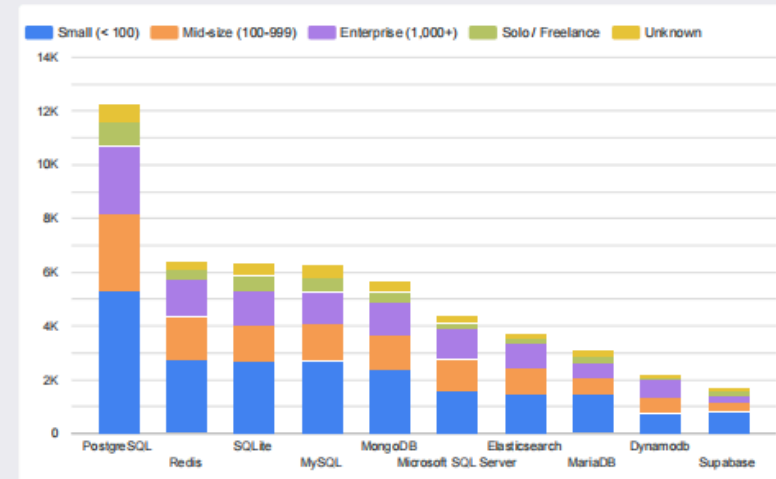
DASHBOARD 2: FUTURE TECH USAGE

Future Technology Trend

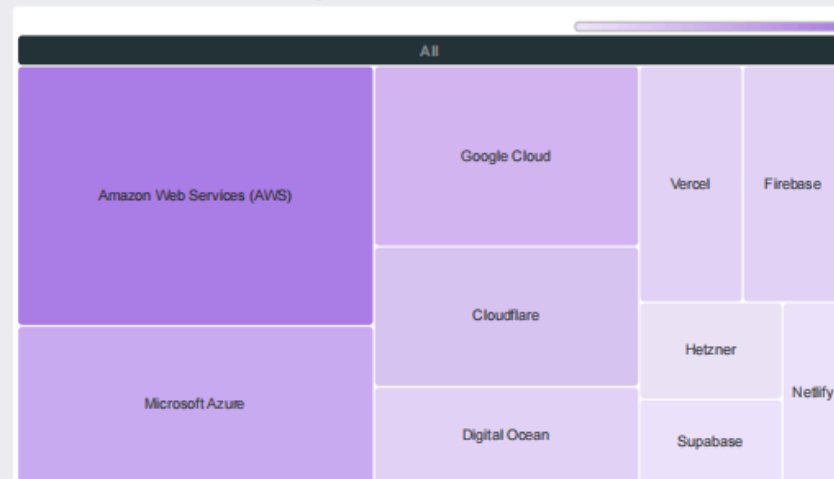
Top 10 Languages Desired



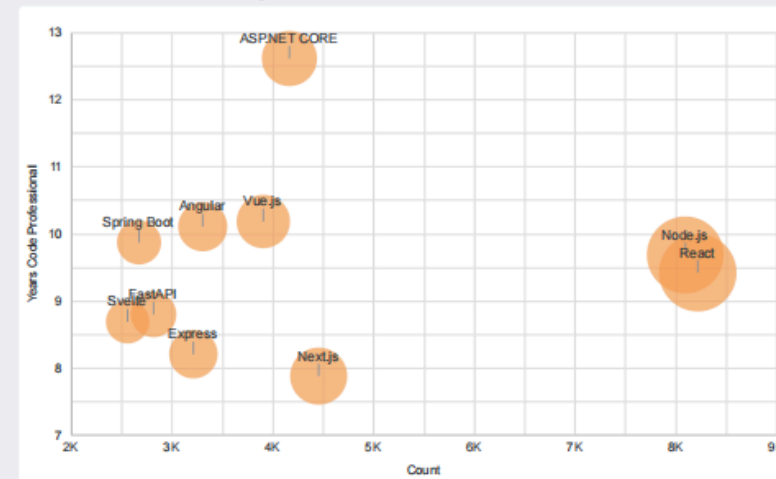
Top 10 Database Desired



Top 10 Platform Desired

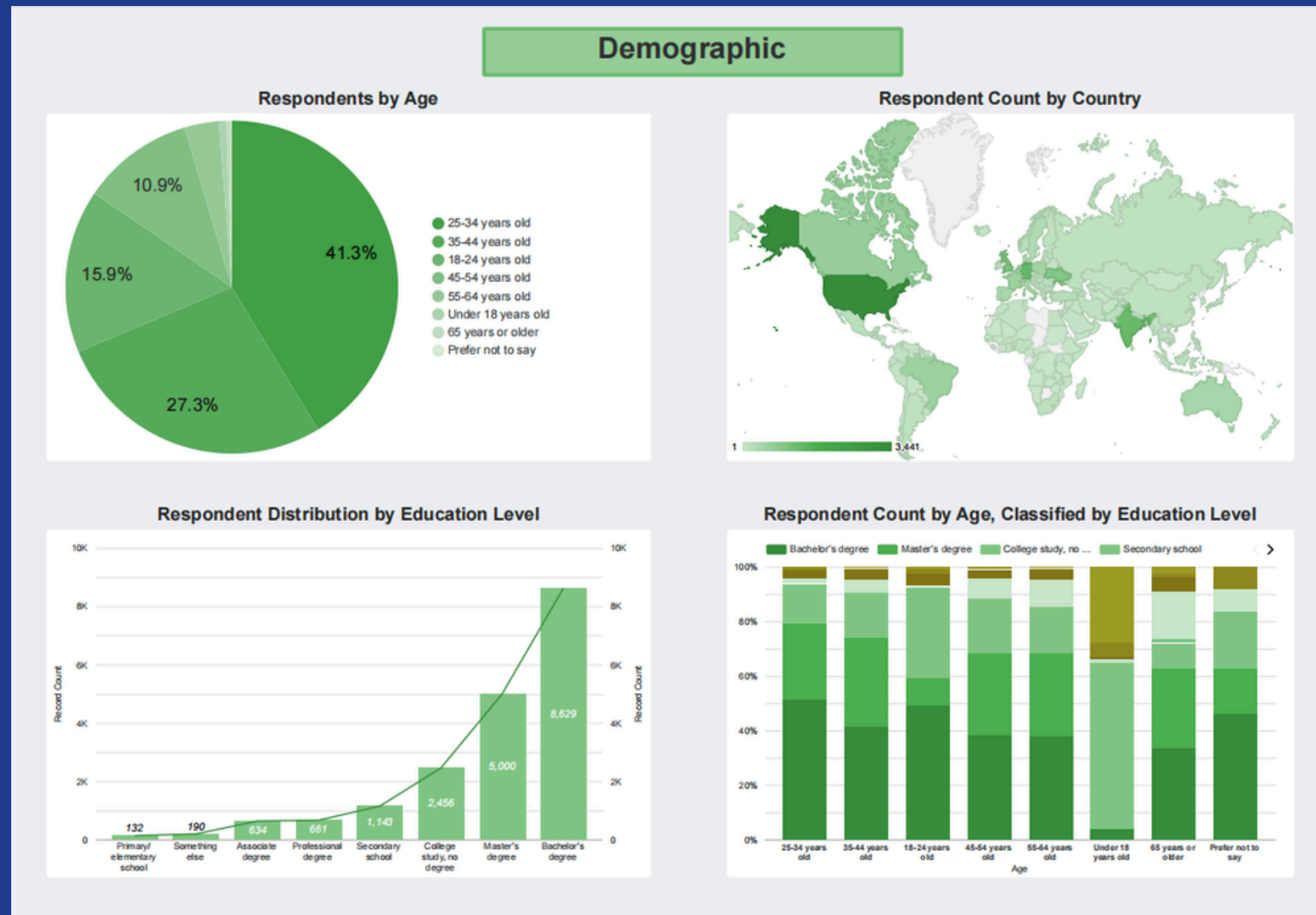


Top 10 Web Frameworks Used



- Go and Rust surge into the top 10 desired languages for next year, displacing legacy scripting tools like PHP and PowerShell
- PostgreSQL expands its lead as the most desired database (~12,200)
- Framework shift toward modern developer experience
- Developer interest in expanding toward specialized edge and deployment platforms like Vercel and Cloudflare

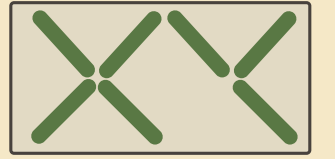
DASHBOARD 3: DEMOGRAPHICS



Young and highly educated workforce demographics

- Early to mid-career:
 - 25-34 (41.3%)
 - 35-44 (27.3%)
 - 18-24 (15.9%)
- Formal university education
 - 8629 respondents have a Bachelor's degree
 - 5000 have a Master's degree and 2456 college level
- United States represents the largest single country market

DISCUSSION



As I am following the guidelines of IBM course, there are many other aspects that I could further take into consideration, such as Job satisfaction. This allows me to further expand this project and dive deeper to gain more impactful insights.

Overall:

- Tech stack continuity: core foundational tools—JavaScript, SQL, Python, PostgreSQL, and AWS continue to be the tools for devs in both the current and future
- Framework selection correlates directly with developer professional experience (YearsCodePro)
 - Seniors (averaging 11.8–12.5 YearsCodePro) - ASP.NET, jQuery
 - Juniors (averaging 8.0–8.5 YearsCodePro) - Next.js, FastAPI
- PostgreSQL serves as the universal database across every business size
- The "Big Three" (AWS, Azure, Google Cloud) handle core compute, while developer-first edge platforms (Vercel, Cloudflare, Supabase) capture rapid deployment workloads

OVERALL FINDINGS & IMPLICATIONS

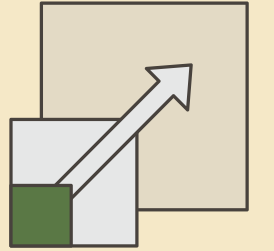
Findings

- PostgreSQL is the #1 database across all organizational scales, while proprietary systems like Oracle and Microsoft SQL Server remain heavily restricted to large enterprises
- Web framework adoption is distinctly segmented by developer experience
- Modern compiled systems languages (Go, Rust) and static typing (TypeScript) represent the fastest-growing desired technologies

Implications

- Engineering organizations should standardize core data storage on PostgreSQL and align backend development with Go, Rust, and TypeScript
- Because the workforce is concentrated in the 18–34 age group, offering structured internal mobility and training in high-demand technologies (Rust, Go, Next.js) serves as a critical lever for engineering talent retention

CONCLUSION



The global developer ecosystem remains supportive, with proven foundational standards (JavaScript, SQL, PostgreSQL) while actively modernizing toward high-performance systems languages (Go, Rust) and agile cloud-native frameworks.

The engineering talent pool is young (68.6% under 35 years old) and academically grounded (72.1% hold university degrees), forming a dynamic and future-ready workforce that promises to drive sustained growth and innovation across the industry.

Therefore, optimize engineering productivity and talent retention by standardizing core development on PostgreSQL and TypeScript/Go, upskilling teams in high-demand technologies like Rust and Next.js, take advantages of cloud infrastructure.

